



Perspective

A perspective on bleb and empty LNP structures

Jens B. Simonsen^{*}*Jbsimonsen Consult, Grønnevej 160, 2830 Virum, Denmark*

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ABSTRACT

Although lipid nanoparticles (LNPs) have been FDA-approved for mRNA delivery, there is still much to learn about these fascinating multi-component delivery systems. Here, I discuss the presence of “bleb” structures on LNPs and the co-existence of mRNA-empty LNPs in LNP-mRNA-based formulations. Specifically, I discuss key articles on these structural and compositional heterogeneities, whether these features present negative or positive LNP attributes, and how to deal with them in research and quality control settings. Additionally, I present current approaches and propose novel strategies on how to study and quantify bleb and empty LNP structures. With the conflicting views on these features in the literature and limited systematic studies on their impact on safety and efficacy, I hope this Perspective will support current and bring forward new thinking about these matters. I anticipate that novel studies and insights could emerge from these lines of thinking, which could potentially enhance the development of safe and efficient LNP-based drug products that will either embrace, leverage, or mitigate the presence of blebs and empty LNPs.

1. Introduction

Lipid nanoparticles (LNPs) have unlocked the vaccine and therapeutic potential of nucleic acids. These breakthroughs include the LNP-siRNA-based drug product Onpattro® for the treatment of transthyretin-induced amyloidosis approved by the Food and Drug Administration (FDA) in 2018 [1] and the two LNP-mRNA-based COVID-19 vaccines from Pfizer/BioNTech (Comirnaty®) [2] and Moderna (Spikevax®) [3] that received full FDA approval in 2021 and 2022, respectively. Today, numerous LNP-mRNA investigational drug products are undergoing clinical trials for various indications, including different types of cancer, cystic fibrosis, and liver-related metabolic diseases to name a few [4]. Current LNP research and development efforts are focused on identifying novel ionizable cationic lipids (hereafter referred to as ionizable lipids) that are combined with helper lipids (phospholipids and cholesterol), and PEG-lipids in different compositional variations [5] and for some approaches, with ligand-conjugated lipids [6]. These efforts aim to customize LNPs for enhanced transfection potency, improved safety, and favorable biodistribution properties [6].

In this Perspective, I discuss the relevance of the physicochemical features of LNPs, which have recently garnered considerable attention within the LNP community. These features include the presence of “bleb” structures on LNPs. Bleb structures, as observed in some LNP-mRNA formulations by cryo-TEM, are protruding structures containing

an aqueous core surrounded by a lipid bilayer [7] (Figs. 1 and 2). The bleb structure is proposed to be phase separated from a solid core, presumably containing ionizable lipids [7]. Additionally, I discuss the co-existence of LNPs containing no mRNA payload (hereafter referred to as empty LNPs) in LNP-mRNA formulations that have been quantified in several studies [8–10].

I present key articles on these two structural and compositional heterogeneities. Interestingly, these studies present divergent conclusions on the impact of blebs on LNPs' transfection potency and the proportion of empty LNPs in LNP-mRNA formulations. Further, I discuss how we should deal with these features, including whether these should be considered good or bad LNP attributes, how to study them, and how research and quality control (QC) laboratories should approach these features.

2. What do we know about blebs on LNPs?

A study by Leung et al. [7] from 2015 used the word bleb to describe liposome-like protruding structures on LNPs. These blebs were observed using cryo-TEM on LNPs containing KC2/DSPC/cholesterol/PEG-c-DMA (50/11.5/37.5/1 mol%) loaded with plasmid DNA or mRNA.

Brader et al. [11] cleverly harnessed the improved cryo-TEM contrast achieved through selective binding of the thionine dye to mRNA. By using this method, they revealed that the location of mRNA in

^{*} Corresponding author.

E-mail address: jbsimonsen@gmail.com.

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bleb-containing LNPs (composition not disclosed) is within the bleb structures (Fig. 1a, top). They also mentioned that “blebs have been widely referred to in the literature as structural defects”. Hence, blebs could be considered a bad attribute.

Henderson et al. [12] evaluated the morphological and functional impact of LNPs dialyzed in phosphate saline (PBS), HEPES saline (HBS), and TRIS saline (TBS) buffers and stored for three weeks at -20°C . All three LNP formulations were based on the standard LNP lipid composition, defined as 50 mol% ionizable lipid, 10 mol% DSPC, 38.5 mol% cholesterol, and 1.5 mol% PEG-lipid using the MC3 ionizable lipid (used in Onpattro®) appeared as solid-spherical particles in cryo-TEM before freezing. Herein, I distinguish solid-spherical LNPs from spherical liposomes with an aqueous core and from non-spherical LNPs, such as bleb-containing LNPs. Storing the LNPs at -20°C led to significant but also distinct changes to the morphology across the LNPs dialyzed against different buffers. While the thawed LNPs in HBS gave rise to the formation of a single bleb structure on all LNPs present in the cryo-TEM image, TBS led to the formation of some LNPs containing more than

one bleb structure in addition to intact solid-spherical LNPs. Lastly, the LNPs in PBS introduced single bleb structures on some of the LNPs as well as liposomes (potentially blebs that have detached from the LNPs). All LNP buffer formulations that were frozen showed significantly reduced in vivo transfection potency four hours after LNP treatment in mice compared to the corresponding non-frozen LNP samples.

A similar study was performed by Meulewaeter et al. [13]. Aiming to improve LNP stability, they evaluated TRIS, phosphate, and PBS dialysis buffers in combination with a continuous freeze-drying step based on spin-freezing. The LNPs were prepared from the standard LNP lipid composition using the C12–200 ionizable lipid. The dialysis of LNPs in TRIS buffer resulted in solid-spherical LNPs, while the additional lyophilizing step introduced some bleb-containing LNPs. Both formulations showed equal transfection potency. PBS dialysis gave rise to many different bleb-structures, and an additional lyophilizing step introduced the formation of multilamellar structures in both the solid-spherical and bleb-containing LNPs. This significantly reduced the transfection potency of the LNPs compared to the non-lyophilized LNP

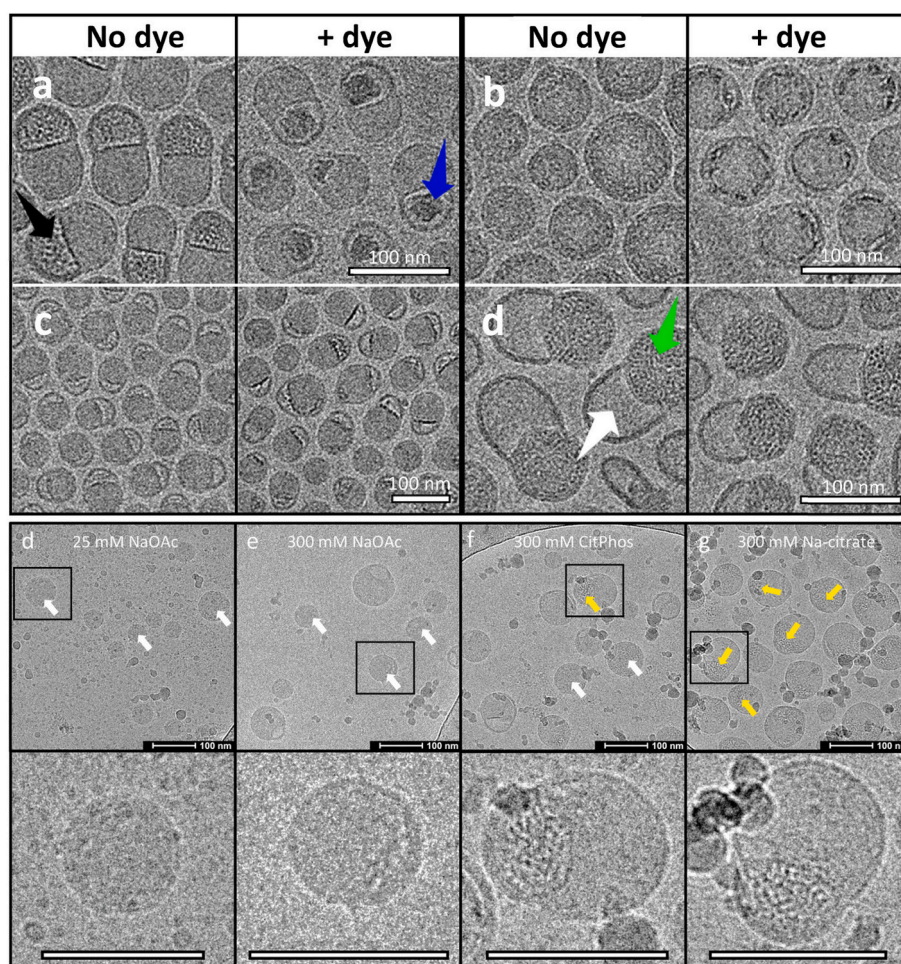


Fig. 1. Top figures: Locating encapsulated mRNA in LNPs by cryo-EM. The effect of thionine on mRNA-LNPs of different morphologies reveals that lipid-dissociated mRNA may reside in bleb compartments (a) or may be more lipid-associated in spherical (b) or less prominently blebbed particles (c). The black arrow in (a) indicates the distinctive mottled mass density of mRNA inside the bleb cavity, which itself is distinguished by a thick, dark periphery. The blue arrow indicates the significant contrast enhancement that occurs when thionine dye is present, thereby identifying mRNA within the bleb. (d) Charge-driven migration of mRNA from the bleb into the body of the LNP. When the mRNA-LNP sample of (a) (no dye) was dialyzed into pH 5 buffer, the images of (d) resulted. The mottled density is now associated with the body of the LNP (green arrow), leaving the bleb cavity devoid of mRNA (white arrow). Addition of thionine to this sample produced no notable contrast change (right), indicating that thionine did not displace lipid. Reprinted figure and figure caption text from [11] (according to Creative Commons CC-BY license). Bottom figures: Cryo-TEM images of LNP luciferase mRNA formulated with various pH 4 buffers: (d) 25 mM NaOAc, (e) 300 mM NaOAc, (f) 300 mM CitPhos (citrate-phosphate), and (g) 300 mM Na-citrate. The white arrows point to LNPs exhibiting normal solid core structure whereas the yellow arrows indicate bleb structures. White scale bar for enlarged images = 100 nm. Reprinted the figure and slightly modified the figure caption text from [14] with permission from John Wiley and Sons in the format of republication via the Copyright Clearance Center. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

formulation. Hence, the studies by Henderson et al. [12] and Meulewaeter et al. [13] collectively highlight that bleb structures induced by freezing and freeze-drying can decrease the transfection potency of LNPs and that the type of buffer has different effects on LNP stability upon exposures to freezing conditions.

In contrast to these studies, Cheng et al. [14] recently showed that an increase in the bleb fraction using the same lipid composition (the standard LNP lipid composition using KC2) but different buffer systems increases the transfection potency of the LNPs in both *in vitro* and *in vivo* studies. Specifically, they show that Na-citrate at 300 mM induced blebs in 59% of the LNPs, while citrate-phosphate and sodium acetate buffers at 300 mM gave rise to lower bleb abundance. 25 mM acetate buffer gave rise to the fewest blebs and exhibited the lowest transfection potency among the investigated LNP preparations. One interesting observation is that there seems to be a positive correlation between the calculated anionic strengths of the buffers and the corresponding bleb fractions, according to my calculations. The lipid bilayer blebs are likely comprised of the lipids known to form liposomal structures such as DSPC, PEG-lipids, and cholesterol. On this note, the study by Leung et al. [7] from 2015 also showed that by substituting the cylindrical-shaped DSPC component (which tends to form liposomal bilayers) with the cone-shaped DOPE lipid, the blebs disappeared and the LNPs became spherical.

Cheng et al. [14] also studied LNPs containing the ionizable lipids MC3, ALC-0315, and SM-102 (the latter two ionizable lipids are used in Comirnaty® and Spikevax®, respectively), but did not study the impact of a freezing step. They propose that blebs enhance the stability of the mRNA payload, thereby increasing their transfection potency. Consequently, they consider blebs an advantageous attribute. A study by Packer et al. [15] highlights that mRNA is prone to form mRNA-ionizable lipid adducts. This observation supports the stabilization of mRNA in bleb structures exhibiting a low abundance of/low exposure to ionizable lipids.

The ways in which the blebs are formed, the different types of blebs, the extend of blebbing (e.g., multiple blebs per LNP and proportion of bleb-containing LNPs), are all factors that may affect the transfection potency. These diverse morphological effects resulting from specific types/concentrations of dialysis buffer and different freezing conditions, underscore the complexity of LNPs and makes it difficult to identify trends and correlations. However, upon a closer examination of the cryo-TEM data reported by Henderson et al. [12], it appears that blebs derived from freezing conditions do not contain (in most cases) the distinctive mottled mass density feature of mRNA in mRNA-containing blebs. In the work presented by Meulewaeter et al. [13] a similar (but less pronounced) observation may be made. Specifically, one out of the two blebbed LNPs seem to be mRNA-empty. These observations presented above indicate that, in order for the blebs to facilitate improved transfection potency/improved mRNA stability, the blebs should contain the mRNA. The formation of the mRNA-empty blebs at freezing conditions/during freezing may be due to the limited mobility of the large mRNA molecule compared to the lipids under such conditions. Along this line, Brader et al. [11] showed that the mRNA localized in blebs is capable of reassociating with the lipidic body of the LNP (in non-freezing conditions), leaving the bleb compartment empty when dialyzed into pH 5 buffer (Fig. 1d, top).

A recent study by Eygeris et al. [16] supports the relation between mRNA-loaded blebs and increased transfection potency. In this study, they compared LNPs loaded with mRNA (prepared by a standard method) using the standard LNP lipid composition based on MC3 (primarily) versus LNPs loaded with mRNA by mixing empty LNPs with mRNA. The latter formulation gave rise to a significant formation of bleb structures containing mRNA (identified in cryo-TEM images, based on the mottled density contrast in the blebs revealed by cryo-TEM). This formulation exhibited an increased transfection potency compared to the classical solid-spherical LNPs prepared by the conventional mRNA-lipid mixing method.

An important point to highlight regarding the proposed positive attribute of mRNA-loaded blebs with respect to increased transfection potency, in contrast to empty blebs, is that we are comparing frozen LNP samples with fresh LNP samples. Hence, other factors during and under freezing conditions than the formation of empty blebs may also contribute to the reduced transfection potency of the frozen LNP formulations, although such factors may likely be highly related to the formation of empty blebs.

In addition to how the lipids are arranged into bleb, solid, lipid monolayer, bilayer, and multilamellar structures, the three-dimensional structure of mRNA is also a topic of interest. The secondary and tertiary structures of mRNA may have an impact on its lipid-mRNA packing, stability, endosomal release efficiency and translation potency. Interestingly, Kloczewiak et al. [17] have shown that thermal unfolding of RNA (T_m) shifts to higher temperatures in LNPs compared to free RNA in solution, indicating enhanced structural stability of the encapsulated RNA. However, structural aspects of mRNA is outside the scope of this Perspective.

Whether blebs are formed and whether they constitute a good or bad LNP attribute may be determined by the overall lipid composition, buffer excipient, the preparation route, and storage. These factors affect the physicochemical and biological properties of the LNPs, including their *in vivo* stability, biodistribution, and transfection potency. Unfortunately, limited data exist on the impact of bleb and empty LNP structures. Regarding the *in vivo* stability of LNP-mRNA particles, it should be recognized that such studies are difficult to perform, due to challenges associated with recovery of pure LNP-mRNA particles (typically, isolated from plasma and serum that contain highly abundant lipid-based biological nanoparticles with similar physicochemical properties as LNPs) [6]. Liu et al. [18] presented cryo-TEM data suggesting that LNP-mRNA particles based on a solid-spherical structure are largely transformed into lipid-bilayer structures after being exposed to rat plasma. Interestingly, some of these structures resemble LNP-mRNA structures that have been subjected to multiple freeze-thaw cycles [11]. In contrast, Francia et al. [19] show that LNPs loaded with magnetic nanoparticles (containing no nucleic acid payload) are structurally intact after high serum exposure. This interesting isolation method (relying on magnetic-trapping of LNPs) enables isolation of pure LNPs from serum with high recovery rate. It remains unclear how the specific payload (magnetic nanoparticles vs mRNA) alters the overall stability of the LNPs in plasma and serum. Having said that, Viger-Gravel et al. [20] have shown based on the standard LNP lipid composition using MC3 that the lipid distributions in the core and at the LNP surface are dependent on whether the payload is siRNA, mRNA, or no payload at all. Note that the varying sizes of these LNPs may contribute to the observed compositional variations, as the surface area-to-volume ratio changes with size. Further, Leung et al. [7] highlight that LNPs containing siRNA produced with an identical lipid composition as their bleb-containing LNP-mRNA particles do not display bleb structures. They suggest that the internal packing of lipids around the shorter siRNA oligonucleotides does not readily lead to segregation of the bilayer-forming DSPC lipid needed to form blebs.

It will be interesting to learn whether the presence of mRNA-containing blebs will increase the *in vivo* stability not only of the mRNA payload but also of the whole LNP-mRNA particle. High *in vivo* integrity of the LNPs may be critical when using ligand-coated LNPs or develop long-circulating LNPs to target extrahepatic cells [6]. It has been shown [21] that LNPs containing a small interior solid core (likely neutral ionizable lipids) situated inside an aqueous compartment bounded by a lipid bilayer (basically, a liposome) (Fig. 2A) can be formed when using a high proportion of phospholipids (40 mol% egg sphingomyelin). Notably, this liposomal LNP structure exhibits extended circulation lifetimes compared to the more conventional spherical LNP-mRNA systems containing 10 mol% DSPC [21]. Hence, there may, at least in this case, be a correlation between circulation time and liposomal-like LNP structures. A “correlation” that may also

translate to the properties of bleb-containing LNPs.

It is striking that the bleb and solid core encapsulated LNP structures may be favorable for extrahepatic delivery and transfection, as these represent a liposomal structure, the ‘mother’ of LNPs.

2.1. Suggestions on how to quantify the proportion of bleb structures beyond cryo-TEM studies

Cryo-TEM is currently the only method that can quantify the proportion of blebs in LNP lipid formulations. Unfortunately, cryo-TEM is not easily accessible to everyone.

If we assume that the lipid composition and arrangement may vary between the bleb and the solid surface due to their different core compositions, surface curvatures, and genesis, then fluorophores that can probe the fluidity, hydrophobicity, or other lipid membrane features may provide semi-quantitative measures of the relative proportions of solid-spherical area or volume and bleb-based LNP surfaces. Such fluorophores could be Laurdan - a membrane fluidity probe, or Nile red - a fluorescent lipophilic dye characterized by a shift of emission from red to yellow according to the degree of hydrophobicity of lipids [24]. It is also worth highlighting novel transmembrane fluorophores [25] - fluorescent molecules that span through the entire lipid bilayer. This type of fluorophore in combination with a lipid fluorophore that incorporates to the outer leaflet of the lipid bilayer may be used to report the proportion of bilayer surface area in blebs relative to the remaining LNP surface area, including the potential solid-spherical lipid monolayer surface (Fig. 2B). However, this approach depends on high bilayer-specificity of the transmembrane fluorophore, limited optical overlap between the fluorophores, the presence of a lipid monolayer (or a lipid surface that cannot host the transmembrane fluorophore) around the solid core of

the bleb-containing LNPs, and that the empty LNPs contain a lipid bilayer surface. Supporting this proposed monolayer model of the solid core surface in blebbed LNPs, the cryo-TEM data presented in Fig. 1 show a thicker surface layer around the blebs compared to the surface layer around the solid compartments. Further, this model is also compatible with the expected physicochemical properties of the hydrophilic outer and inner surfaces of the proposed phospholipid-rich bilayer in blebs, in contrast to the proposed phospholipid-rich lipid monolayer that is only hydrophilic at the outer surface and hydrophobic at the inner surface, which interfaces with the hydrophobic solid core. A back-of-the-envelope calculation of the lipid surface thickness, including all DSPC and DMG-PEG2000, as well as 50% and 10% of the cholesterol and MC3 content, respectively, yields a thickness of approximately 5 nm, which corresponds to the thickness of the DSPC lipid bilayer in an unilamellar liposome [26]. These calculations account for: (i) the molecular weight, bulk molar proportion (using the standard LNP lipid composition), and the partly known density of each of the lipids, including MC3 (0.89 g/ml), DSPC (1.0 g/ml), cholesterol (1.07 g/ml) and DMG-PEG (1.0 g/ml), (ii) the 1921 nucleotide-long luciferase mRNA (frequently used) with an expected density of 1.62 g/ml [27], (iii) an N/P ratio (the number of amino groups from the ionizable lipids per phosphate group on the mRNA) of 6, (iv) an LNP size of 80 nm (excluding the PEG-layer), and (v) excluding water in the LNP (a poor approximation – but it is difficult to provide a qualified guess). This simple calculation (based on calculating the volume expanded by the ‘surface lipids’ relative to the total volume and converting that number to a thickness) aims to highlight that if the DSPC and DMG-PEG lipids form a lipid bilayer bleb, then a limited amount of DSPC lipids would be available to cover the remaining solid core with a DSPC-containing bilayer. However, the type of lipid layer (monolayer, bilayer, or

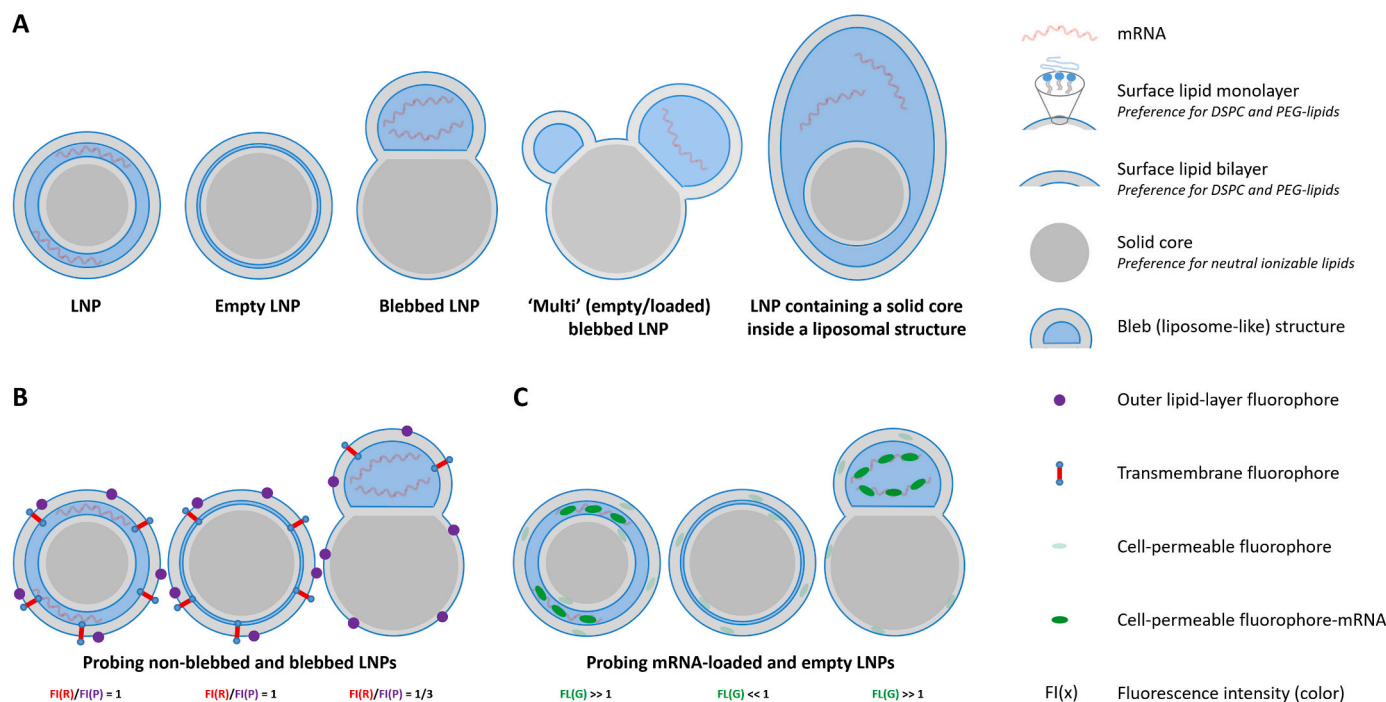


Fig. 2. A: Simple illustrations of the key LNP structures discussed in this Perspective. These structures include spherical loaded and empty LNPs, single and ‘multi’ bleb-containing LNPs, and the solid core encapsulated liposome-like LNP. Many models have been proposed for the spherical nucleic acid-containing LNPs [22]. Here, I use a bilayer/monolayer model for simplicity and for the scientific reasons mentioned in the main text below. The illustration of the surface lipid monolayer highlights two DSPC molecules and a PEG-lipid that are localized preferentially at the LNP surface relative to the core [23]. B: Proposed strategy to probe the bleb versus solid-spherical surface area by using the ratio of the fluorescence intensity from a post-inserted transmembrane fluorophore and the fluorescence intensity from a fluorophore post-inserted into the outer lipid layer. This analytical method should be applicable for both bulk measurements and single-particle analysis. C: To quantify the proportion of mRNA-loaded LNPs using a high-sensitivity flow cytometer (NanoFCM), Geng et al. [9] propose to label mRNA in mRNA-loaded LNPs with a cell-permeable nucleic acid-sensitive fluorophore (they use SYTO 13). The relative fluorescence intensities are shown here to illustrate how this information can be used to identify bleb and empty LNP structures. To keep the illustrations simple, the classical PEG-coating has been omitted.

multilamellar) present around the solid compartment in blebbed LNPs depends on the lipid composition, buffer, LNP formation and storing conditions, and the shape and amphiphilicity of the ionizable lipid in its neutral state. On this note, Meulewaeter et al. [13] observed multilamellar structures in a lyophilized LNP formulation, both in solid-spherical and bleb-containing LNP structures using the highly branched C12–200 ionizable lipid.

While the approaches for probing bleb structures and the monolayer model proposed above are not applicable to all types of LNPs, their main purpose is to encourage considerations of how fluorophores may serve as tools for identifying blebs and for developing our understanding of how lipids can arrange at the surface of LNPs. A universal method for evaluating bleb structures with post-inserted fluorophores may be unrealistic because of the high heterogeneity across LNP formulations. Nevertheless, a combination of fluorophores that can interrogate various lipid features may facilitate the development of valuable qualitative assays for specific types of LNPs. One such assay might be geared to determine whether a significant proportion of blebs is present in an LNP formulation post-formation or during storage under freezing conditions. The simplest assays would rely on bulk-based fluorescence measurements. However, a more valuable assessment of LNP-blebs is likely obtained from single-particle analysis that avoids averaging out key information about ratios between different fluorophores on single LNPs in heterogeneous LNP samples. Especially in cases where both surface monolayer and bilayer LNPs coexist in the sample. Importantly, titration studies will have to be performed to evaluate whether fluorescent lipid probes can be used for bleb identification studies.

In addition to these suggested fluorescence measurements, super-resolution fluorescence microscopy studies of single particles may reveal the presence of asymmetric fluorophore probe distributions across the individual bleb-containing Janus particle-like LNPs. Likely, the larger-sized LNPs are the only ones from which asymmetries would be observable. In such studies, some of the post-fluorophore-labeling strategies discussed above could be tested too. Studies of the coexistence of different lipid domains within the same vesicle have already been performed [28], albeit focusing on giant (micrometer sized) vesicles that are much larger than LNPs.

Fluorescence microscopy studies have revealed that the lipid composition heterogeneity, across individual liposomes in the size range ~ 50 to ~500 nm [29], is far larger than expected from a binomial distribution. The complement and antibody opsonization binding to liposomes is also highly heterogeneous across individual liposomes [30]. Likely, this is a result of the compositional heterogeneity. These large compositional heterogeneities observed for liposomes may also be present in LNPs and drive the different heterogeneities present in some LNP formulations, including empty/loaded and bleb/spherical LNP structures. Notably, a higher compositional heterogeneity across individual particles is expected for the multi-component LNPs compared to the less compositionally and structurally complex liposomes. Lipid compositional heterogeneities across individual lipid-based particles have also been reported by Barriga et al. [31] using SPARTA, a single-particle Raman spectroscopy technique. While Li and Hu et al. [32] reported on compositional heterogeneities in LNPs containing fluorophore-labeled surface lipids and siRNA using cylindrical illumination confocal spectroscopy. For such studies, it is important to quantify and account for the measured variations caused by the measurement equipment, e.g., laser power fluctuations and Poisson-associated detector variations, and low signal levels [33], to derive the actual lipid and mRNA compositional heterogeneities across individual LNPs in an LNP sample.

Returning to the fluorescence-based methods, some precaution should be exercised when using fluorophores for quantitative and qualitative studies of nanomedicines [34], including LNPs, because optical changes of the fluorophore induced by the surroundings can both be exploited but also lead to unforeseen changes that are difficult to interpret. Further, interactions between fluorophores can also alter the

optical properties of the fluorophores. Such interactions include fluorophore association within the LNP lipid layers or in solution, leading to the formation of dye-nanoparticles, resulting in fluorescence quenching (including FRET quenching in case two different fluorophores with optical overlap are being used). These and other effects that can alter the optical properties of the fluorophore are discussed in [34].

The ideal fluorophore for the assessment of e.g. the surface area of LNPs should not form aggregates in solutions with LNPs but associate uniformly with the LNP lipid surface in a wide dye:lipid ratio range without causing particle disruption and flip-flop between lipid layers. Keep in mind that fluorophores used to stain lipid membranes are amphiphilic and therefore tend to form micellar and liposomal structures.

NMR methods may also allow to distinguish between bleb and solid LNP structures. This technique can provide information about the mobility of lipid moieties (atoms) that may differ in a bleb-bilayer compared to the lipid surface around the solid LNP core. For example, ^1H magic-angle spinning NMR has been used to quantify the fraction of lipids in the disordered state in vesicles as a function of temperature. Here, the results were in good agreement with the area fraction of disordered domains in giant unilamellar vesicles of the same lipid composition determined by fluorescence microscopy [35]. Interestingly, Wang et al. [36] highlight that surface structural changes of the PEG-lipid ALC-0159 used in Comirnaty® detected by 1D ^1H NMR likely correlated with elevated bleb formation on aged LNPs observed with cryo-EM.

It will be interesting to see if simple and robust assays (not relying on cryo-TEM), can be developed to assess the proportion (area or volume) of bleb structures in LNP-mRNA formulations.

3. What about the empty LNPs?

A study by Li and Hu et al. [8] reported the existence of 40–80% empty LNPs in LNP-mRNA formulations based on the standard LNP lipid composition (using MC3 and N/P ratios ranging from 1 to 12). They used a single-particle fluorescence method to quantify the proportion of empty LNPs. Although a small fraction of empty LNPs is always expected, the proportion of empty LNPs reported by Li and Hu et al. seem surprisingly high. This is because, a Poisson distribution that assumes no preferential binding between the LNP components, equally sized LNPs, and an average of two mRNA molecules per LNP gives a percentage of empty LNPs of 13.5%. Li and Hu et al. reported 77% of empty LNPs (Fig. 3A) for LNPs with an average copy number of two mRNA molecules per LNP based on an N/P ratio of 6. However, a Poisson distribution is not descriptive, because there are different types of interactions involved in LNP self-assembly and because LNPs show some degree of polydispersity. Additionally, the higher diffusion of lipids compared to the large mRNA molecules during mixing could, under certain formulation conditions (e.g., when the mixing is determined by diffusion rather than turbulent mixing) favor the formation of empty LNPs. Kulkarni et al. [37] have suggested that large payloads promote the formation of a significant population of empty LNPs. Münter et al. [10] recently supported this statement by comparing the proportions of empty LNPs formed with siRNA, mRNA, and plasmid DNA payloads.

One concern related to the interesting study by Li and Hu et al. [8] is their use of highly Cy5-modified mRNAs (about 40 Cy5-modifications on the 1929 nucleotide long luciferase mRNA were present in this study according to the first author). This significant mRNA modification is important for mRNA tracking but may alter the physicochemical properties of the mRNA that drives the encapsulation process. However, Li and Hu et al. do report that the Cy5-mRNA modification did not perceptibly alter the size, morphology, surface charge, and encapsulation efficiency of the LNPs compared to the non-labeled LNP-mRNA formulation. Further, the proportion of empty LNPs reported may be sensitive to the choice of threshold used to define an LNP in this setup.

Interestingly, a recent study by Münter et al. [10] used a fluorescence

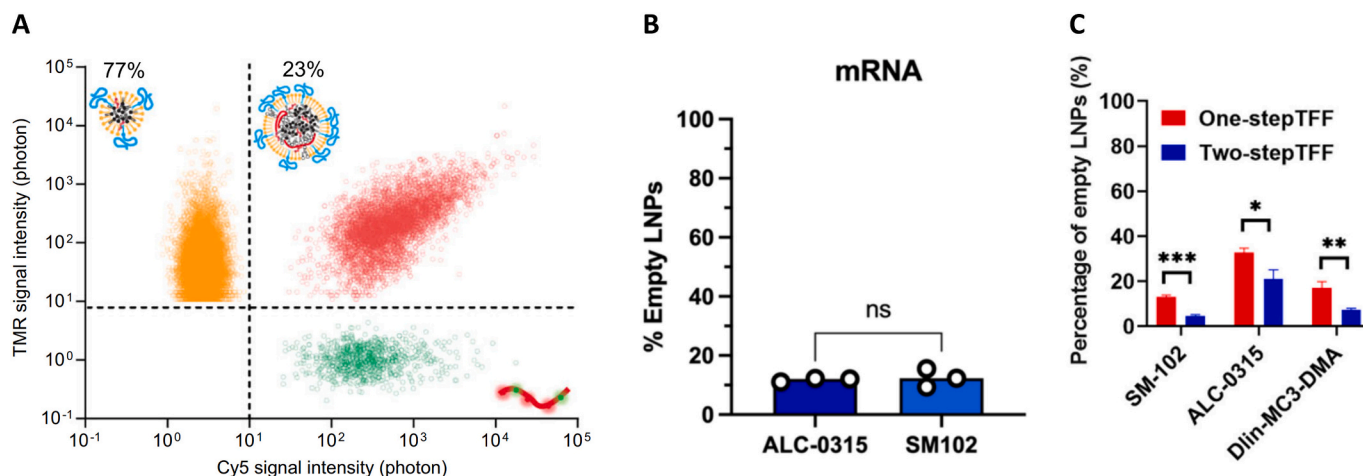


Fig. 3. Reported percentages of empty LNPs across different LNP formulations and techniques discussed herein. **A:** The cylindrical illumination confocal spectroscopy-based study focused on the standard LNP lipid composition using MC3 reports 77% empty LNPs when using an N/P ratio of 6, Cy5-labeled mRNA, and TMR-labeled lipids. The figure is reused from Li and Hu et al. [8] published by Nature Portfolio under the Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>). **B:** A fluorescence microscopy-based study highlighting that LNP mimics of Comirnaty® (based on ALC-0315) and Spikevax® (based on the standard LNP lipid composition including SM-102) contain less than ~16% empty LNPs. The figure is reused from Münter et al. [10] published by Elsevier under the Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>). **C:** A flow cytometry (NanoFCM) based study using post-labeled mRNA (as opposed to pre-labeled mRNA used in the studies shown in A and B). In this study LNPs containing the standard LNP lipid composition and SM-102, ALC-0315, and MC3 (Dlin-MC3-DMA) were evaluated in combination with a one-step or two-step tangential flow filtration. Reprinted figure from Geng et al. [9] with permission from Elsevier in the format of republication via Copyright Clearance Center.

microscopy assay to observe a much lower percentage of empty LNPs (average $12.1 \pm 2.0\%$, Fig. 3B) compared to Li and Hu et al. [8]. This type of assay was also used to assess the lipid and opsonization heterogeneities of liposomes highlighted above [29,30]. In this new study, Münter et al. evaluate dual-labeled (lipid and mRNA) LNPs mimicking Spikevax® and Comirnaty® lipid-wise, using a 996-nucleotide mRNA. They also show that the method used to prepare the LNPs has a significant impact on the proportion of empty LNPs that are being formed. The fraction of empty LNPs increased about three-fold when the LNPs were prepared by pipette mixing compared to microfluidic mixing. One potential challenge with this great assay is that the smallest LNPs may not be efficiently coupled to the streptavidin-coated surface, as these have a high surface curvature and limited numbers of biotin-lipids to facilitate the surface immobilization of the LNPs. Hence, this assay may underestimate the fraction of empty LNPs as this size population likely contains a large proportion of empty LNPs. Münter et al. also evaluated the fraction of empty LNPs based on LNPs containing the standard LNP lipid composition with C12–200, the helper lipid DOPE instead of DSPC, and siRNA, mRNA, and plasmid DNA payloads. Notably, they observed a positive correlation between payload size and the fraction of empty LNPs.

Geng et al. [9] found 13.0%, 17.1%, and 32.7% empty LNPs using the standard LNP lipid composition with SM-102, MC3, and ALC-0315, respectively. They also demonstrated a significant decrease in these numbers after introducing a two-step (instead of a one-step) tangential-flow filtration following lipid-mRNA mixing. This study used post-labeling of mRNA with SYTO 13 as opposed to the pre-labeling based on Cy5 conjugation to mRNA used by Li and Hu et al. [8] and Münter et al. [10]. The detection method used by Geng et al. relies on a high-sensitivity flow cytometer (NanoFCM). However, their post-labeling approach may be problematic, as cell-permeable molecules are amphipathic and could thus bind to empty LNPs (Fig. 2C). Importantly, these dyes typically show a large fluorescence enhancement upon binding mRNA (Fig. 2C). On this note, preliminary studies conducted by the author of this Perspective and others using FIDA (a technique based on Taylor's dispersion), reveal that SYTO 13 (added to LNPs at room temperature prior to measurement) does not produce fluorescence in empty LNPs with the standard LNP lipid composition and SM-102

(without mRNA). In contrast, LNPs containing mRNA present a pronounced fluorescence signal. When using post-staining of encapsulated mRNA, it may be time consuming to establish (likely for each unique type of LNP) the optimal condition dye/mRNA ratio and incubation conditions for sound quantification of the fraction of empty LNPs. Nevertheless, this strategy is indeed very rewarding if successful. Interestingly, NanoFCM has also been used to quantify the mRNA encapsulation efficiency (data not published). It is however, a requirement for such measurement to be valid that the mRNA stain maintains binding properties and optical properties for mRNA inside an LNP and for free mRNA in solution. Whether these requirements can be met may depend on the location of the mRNA within the LNP.

These various studies reporting different fractions of empty LNPs collectively support that factors such as lipid composition, mRNA payload size, preparation method, and analysis technique impact the fraction of empty LNPs [8–10].

The removal of either low, medium, or high fractions of empty LNPs may likely improve both the transfection potency per LNP and the safety profile of the LNP formulation. Hence, the empty LNPs could be considered a bad feature. However, the addition of a purification step (e.g., by density gradient centrifugation) to remove or reduce the fraction of empty LNPs, would complicate manufacturing of the LNP product. Further, it is also questionable how efficient density gradient ultracentrifugation is for such applications [38]. From a research and development perspective, the inclusion of an isolation step for analytical purposes could potentially provide guidance on the lipid composition of the mRNA-loaded LNPs, which may differ as highlighted by Viger-Gravel et al. [13] and potentially be better for mRNA encapsulation, compared to empty LNPs. This 'better' LNP formulation/lipid composition may, however, form new LNP heterogeneities, including empty LNPs that could lead to an overall less potent and less favorable bio-distribution profile. Further, we cannot rule out that empty LNPs present in the LNP-mRNA formulation could have a different *in vivo* behavior than the mRNA-loaded LNPs that could turn out beneficial for the biological impact of the mRNA-loaded LNPs. Specifically, the empty LNPs may have a preference for the phagocytic cells and keep these cells occupied due to their potentially different lipid composition at the surface and morphology compared to the loaded LNPs [7,20]. This could

improve the non-phagocytic mRNA delivery. In such highly speculative scenarios, the empty LNPs could play a positive role. A further challenge to understanding the significance of empty LNPs (e.g., by performing empty LNP spike-in experiments) is that 100% empty LNPs prepared without mRNA may behave slightly different from the empty LNPs co-existing with the mRNA-loaded LNPs due to the compositional and structural reasons discussed above.

To increase the fraction of mRNA-loaded LNPs during preparation, it seems appropriate to consider using either a smaller payload (if possible), optimizing the N/P ratio, and/or lowering the ionic strength of the buffer during the lipid-mRNA mixing. The latter approach will increase the electrostatic interactions between the positively charged ionizable lipids and the negatively charged backbone of mRNA. Furthermore, applying methods that ensure proper mixing and dialysis also reduce the fraction of empty LNPs in the final LNP formulation [9,10].

3.1. Alternative methods to quantify the fraction of empty LNPs

Cryo-TEM studies based on thionine-staining may be used to quantify the fraction of empty LNPs. The cryo-TEM study focused on blebs by Cheng et al. [14] report that more than 60% of the LNPs contain one or more mRNA. This study uses the measured bleb area (which correlates well with the theoretical area filled up by one, two or more mRNA molecules) to estimate the copy number of mRNA per LNP. However, for both of these cryo-TEM approaches, it is uncertain whether a proportion of the mRNA is localized in the solid core and thus inaccessible to either the thionine staining or the blebs. In the event that the core consists of wax-like non-charged ionizable lipids (this is likely the case if the ionizable lipids do not contain a significant amount of hydrophilic functionalities), such a hydrophobic environment will be incompatible with the presence of the hydrophilic mRNA cargo if not trapped during the formulation process. Moreover, the study by Brader et al. [11] also shows that in spherical-solid LNPs, the mRNA is present below the LNP lipid surface (Fig. 1b, top). It should also be noted that bleb-containing LNPs without mRNA (i.e., without the distinctive mottled mass density feature of mRNA inside the bleb cavity in cryo-TEM) have been reported in fresh LNP-mRNA formulations [36].

Super-resolution microscopy may also be used to quantify the fraction of empty LNPs using either pre- or post-labeling mRNA strategies. Importantly, super-resolution microscopy enables single fluorophore detection, which relaxes demands on the degree of mRNA-labeling.

Single-particle Raman spectroscopy techniques may also be suitable to identify mRNA-loaded LNPs based upon the label-free readouts from mRNA and lipid active Raman signals from individual LNPs. Alterations including damping of the vibrational states that are probed by Raman spectroscopy induced by the local environment could potentially pose some challenges to the quantitative information derived from this method. However, the label-free feature of e.g., Single Particle Automated Raman Trapping Analysis (SPARTA) techniques is indeed very encouraging.

Bulk-based methods that have been suggested to provide estimation of the proportion of empty LNPs include analytical density matching [39] and density gradient [38] ultracentrifugation methods. To estimate the fraction of empty LNPs with such bulk-based studies, some normalization with respect to particle number, lipid concentration, or lipid mass is needed. The question is whether LNPs loaded with a single mRNA molecule can be completely isolated from the slightly less dense fraction containing the empty LNPs. This will be particularly challenging for large LNPs containing a single small mRNA molecule that may have a lower density than the smallest empty LNPs.

4. The inherent complexity of LNP-mRNA particles and tools to study LNPs

In general, these structural and compositional heterogeneities add to

the overall complexity of LNPs, which typically contain four different lipids (ionizable lipids, phospholipids, cholesterol, and PEG-lipids) [5] and one or two types of nucleic acid payloads. We may need to recognize that this multi-component swiftly shaken cocktail made at room temperature - the LNPs - may not necessarily reach a thermodynamic equilibrated state. Instead, the LNPs may be trapped in a meta-stable state or states. Thus, small changes in composition, preparation route, storage, and pre-analysis handling can lead to great structural changes. Along these lines, Hald et al. [40] showed that rapid pipette mixing of the standard LNP lipid composition using MC3 and mRNA gave rise to LNPs with a size consistently around 130 nm, independent of the hands carrying out the pipetting, while the identical composition gave rise to ~85 nm LNPs when using a commercial microfluidic mixing system (NanoAssemblr™). This observation is in accordance with the impact of the flow rate on LNP size when prepared with microfluidic systems [41], at least when standard lipids and compositions are used. Likewise, the morphological remodulations of LNPs introduced by freezing [12] and lyophilization [13] underscore the sensitivity of lipid-packing/state with respect to temperature.

With these considerations in mind, a big question for the field remains whether we should embrace the LNP compositional and structural heterogeneities and try to optimize and take advantage of these features or whether LNP research and development programs should aim to eliminate the formation of bleb structures and empty LNPs (if possible at all). Interestingly, bleb structures have been referred to as ‘artifacts’ and considered bad in opposition to Cheng et al. [14] and Eygeris et al. [16] findings, while mRNA-empty LNPs likely lower the therapeutic window of the LNP drug product and hence are a bad LNP attribute. Furthermore, if we gain a sufficient understanding of whether bleb and empty LNP structures are positive or negative LNP attributes, should research articles on LNPs and quality control (QC) analysis of LNP-based drug products then in addition to size, polydispersity index (PDI), zeta potential, encapsulation efficiency, and composition also report on the proportion of bleb structures and empty LNPs? If so, which methods should be used? However, it is not feasible to mandate that researchers acquire occasionally costly research equipment. Further, it is challenging to incorporate these novel technologies in QC laboratories where e.g., a new software update can require months of testing and reporting and the need for manual tool optimizations and potential clotting issues are incompatible with strict QC procedures that often rely on robust and simple workflows.

In research and QC, dynamic light scattering (DLS) is without question the gold standard method for determining the average size, PDI, and the zeta potential of nanomedicines. These characterization parameters are also the most reported parameters in the LNP literature in addition to the nucleic acid encapsulation efficiency, and mRNA concentration. However, DLS has its challenges and limitations [42], and FDA prefers that the average nanoparticle size is defined based on volume or mass if particle density is known [43]. This can be difficult to satisfy with the light scattering intensity weighted readouts from DLS measurements. Fortunately, many new and interesting commercial tools have emerged, based on high-sensitivity flow cytometry [9]; fluorescence microscopy [44]; nanoparticle tracking analysis (NTA) [45]; optical nanoparticle trapping combined with Raman spectroscopy [46]; and microfluidic systems based on Coulter principles [47] or based on flow-induced dispersion principles [48]. These tools can in addition to average size and PDI assessments provide information about the particle concentration of LNPs, encapsulation efficiency, proportion of mRNA-loaded LNPs, and (to some extent) assess the protein corona formed on LNPs directly in biologically relevant matrices such as plasma and serum.

The more well-established methods, such as small-angle X-ray and neutron scattering techniques (SAXS and SANS) can provide direct information about whether ordered lipid or lipid-mRNA structures are present in LNPs and whether changes in the arrangement of the LNP components occur as a function of physicochemical alterations of the

LNP formulation. However, they are not optimal tools for studying highly heterogeneous LNP samples. This is because they are bulk-based methods and that limited numbers of independent parameters to describe the system(s) can be derived with these methods without avoiding overfitting [49]. Although SAXS has been used to identify different conformations of proteins and protein oligomers in one sample, such samples are still considered fairly homogenous (compared to a heterogeneous LNP sample) due to their well-defined length (primary sequence) and scattering cross section. To make SAXS and SANS useful for describing heterogeneous LNP samples may require detailed molecular and LNP structural constraints derived from complementary analysis.

Molecular dynamics (MD) simulations [23,50] that like SANS [51,52] have been used to estimate the location of the different types of lipids in LNPs (core vs shell) may also fall short in describing heterogeneous LNP systems. Specifically, it is questionable whether MD methods are capable of identifying the (potentially) trapped meta-stable LNP structures that are formed under complex reaction conditions. Along this line, the MD method needs an appropriate initial position of the LNP components, and limited time frames can be modeled with this CPU-expensive method. Considerable efforts are being devoted beyond increased CPU power to improve the modeling of large systems over longer time frames, including metadynamics, replica exchange, umbrella sampling, Markov state models and coarse-graining methods [53,54].

Overall, the combination of newer single-particle tools and cryo-TEM enables detailed characterization of structural and compositional heterogeneities in LNPs. Thus, we have a foundation to correlate these features with their *in vitro* and *in vivo* biological performances.

5. How research and QC laboratories should approach bleb and empty LNP structures

On one hand, LNP-mRNA formulations exhibit highly complex compositions and structures. On the other hand, clinically staged LNP-mRNA products have demonstrated reproducibility in their production processes. Hence, we might argue that the current critical quality attributes (CQAs), such as LNP average size, PDI, encapsulation efficiency, mRNA concentration, and lipid composition, are sufficient to ensure the proper quality of a final drug product, particular concerning the physicochemical properties of the LNPs. We might also argue that the presence of bleb structures and empty LNPs is directly related to the lipid and RNA composition and preparation route, including process-related factors. Hence, information about these latter parameters remain the most critical (need to know) attributes. Therefore, in some cases, it becomes difficult to claim that empty and bleb-containing LNPs are bad attributes for a given LNP formulation, as they constitute an integral part of the LNP formulation that may be efficacious and relatively safe.

Considering the available literature and appreciating the propensity for heterogeneity in LNP samples of complex compositional formulations formed in a rapid fashion, it becomes evident that both empty LNPs and bleb structures are present in many LNP-mRNA formulations. From a QC perspective, this observation may not necessitate radical changes in the CQAs reported for LNP-mRNA drug products, as long as the conventional CQAs are met. However, from a research and development point of view, these additional complexities and heterogeneities should be studied in more detail. Serving this purpose, several new types of equipment are capable of complex interrogations, as briefly mentioned above.

6. The relevance of LNP reference materials and cross-equipment and laboratory comparison activities

Recently, the National Institute of Standards and Technology (NIST) has initiated an effort to develop an LNP-RNA Research Grade Test Material (RGTM) [55] that will facilitate better comparison studies

across different LNP characterization tools. One major concern when comparing data of the proposedly ‘same’ LNPs from different laboratories using various characterization methods is that the quality and identity of the LNPs are likely different across the laboratories due to slight variations in LNP composition, preparation, and/or storage. NIST’s LNP-mRNA RGTM effort will indeed put the LNP field in a better position to understand the abundancies of structural and compositional heterogeneities, as well as the limitations and advantages of the many different tools currently used to characterize LNPs. On this note, a recent study by Parot et al. [56] compared different methods to assess LNP size and other quality attributes based on LNPs made from the standard LNP lipid composition containing either MC3 or SM-102. Parot et al. show that orthogonal methods (methods that provide the same measure but rely on a different measuring principle) produce different outcomes. Their study includes discussions of measurements that can provide information about the structure/morphology of LNPs, including asymmetrical flow field-flow fractionation (AF4)-multi-angle light scattering (MALS)-DLS, analytical ultracentrifugation (AUC), and differential scanning calorimetry (DSC). Specifically, these methods can be used to derive information about LNP morphology, LNP density, and lipid-lipid and lipid-mRNA packing (T_m values). Lastly, Parot et al. encourage international collaborations toward the development of measurements and standards for LNP-RNA systems, which are in line with the NIST effort and similar requests coming from both regulators [57,58] and policymakers [59].

7. Summary

Established and emerging tools for analysis of nanoparticles have brought to light the complexity and heterogeneity of LNPs, including the presence of blebs and empty LNPs. These findings have to some extent added more questions and more confusion to how we should design potent and safe LNPs. However, it makes sense that uncovering the landscape of LNP structures is an activity that goes prior to and happens faster than deriving structure-function relations.

I find that it is important to stress the dependency of nucleic acid, lipid, and additional excipient composition and process-related factors on compositional and structural heterogeneities in LNPs. This means that a given reproducible LNP formulation with an overall attractive efficacy and safety profile is considered good, no matter whether it contains a mixture of empty, loaded, solid-spherical, and bleb-containing LNP structures. This should be central when identifying CQAs that are robust for a given LNP-mRNA drug product. In other words, it should not be a goal in itself to identify good and bad sub-populations within an LNP formulation, if these cannot be efficiently isolated or removed, as each of them is a product of the bulk composition and preparation route.

To enhance our understanding of various novel and established tools for accurate characterization of LNPs, including their bleb and empty structures, as well as other features, I greatly appreciate NIST’s efforts in developing standard LNP materials and similar initiatives [56].

Although I have not definitively answered the question of whether bleb and empty LNP structures are positive or negative attributes, I believe I have offered literature overview, ideas, and new perspectives on these matters. While some of my speculations may be contradictory, their purpose is to broaden our awareness of potential challenges and opportunities in the LNP field. I hope that the presented information can guide further studies, helping us to better evaluate and understand the value of these attributes. As part of this, others and I propose to explore different types of stains/fluorophores in different combinations to gain insights about LNPs, including their fraction of empty LNPs, size, blebs, internal structure, aggregates, and particle concentration when combined with fluorescence-based single-particle analysis tools. Such measurements would however require well-defined controls, careful titration and incubation optimization studies, and awareness of optical overlap between fluorophores as well as self-quenching and other effects

[34] to avoid drawing wrong conclusions.

The conflicting views based on limited studies on these matters make it difficult to provide definitive answers. Maybe the complexity of the LNP lipid and payload compositions and preparation routes drives the observed differences and thus highlights that blebs and empty LNPs can be either good or bad depending on the LNP composition, process-related factors, storage condition, and the biological target.

That said, if feasible, limiting the proportion of empty LNPs without altering the physicochemical properties of the mRNA-loaded LNPs would likely enhance the therapeutic window for the majority of LNP formulations. Further, the sparse data on the impact of bleb structures on the transfection potency of LNPs indicate that blebs containing mRNA compared to empty ones are preferable when compared to their solid-spherical counterparts. This general observation relies, however, on comparing frozen LNP samples (containing primarily mRNA-empty blebs) with fresh LNP samples containing mRNA-loaded blebs.

Drug development programs tend to adhere to simple and uniform formulations, aiming for easier regulatory and manufacturing pathways. However, it would be intriguing to explore the complex structural features of LNP-mRNA particles such as the bleb structures and determine how they may be tuned/controlled as attributes that give rise to attractive biodistribution, potency, and safety profiles. As part of this activity, we need to understand their potential impact on immunogenicity (including the immune response toward the asymmetric-shaped LNPs containing blebs and their Janus-like particle appearance) and whether the increased transfection potency is solely determined by mRNA stability or whether the blebs also lead to increased endosomal escape of the blebs or the entire LNP.

I hope that future structure-function studies will guide the development of safe and efficient LNP-based drug products that will either embrace, leverage, or mitigate the presence of blebs and empty LNPs.

CRedit authorship contribution statement

Jens B. Simonsen: Writing – original draft, Conceptualization.

Declaration of competing interest

None.

Data availability

My commentary show data that is already published by others. I have received copyright permissions to reproduce these figures.

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